Lecture 1
Course Introduction

Prof. Yinqian Zhang

Spring 2025

General Course Info

- Course Name: **(Introduction to) Operating Systems**
- Term: Spring 2025
- Time: Tuesday. 14:00-15:50
- Location: **TBD**
- Lecturer: Prof. Yinqian ZHANG (yinqianz@acm.org)
- Lab instructor:
  - Prof. Yinqian ZHANG (yinqianz@acm.org)
  - Ms. Yun SHEN (sheny@mail.sustech.edu.cn)

Self Introduction

Open Office Hour:
Wed. 9:00 ~ 11:00
615 South CoE Building

- A **professor**
  - of Computer Science and Engineering at SUSTech
  - Previously at Ohio State University in USA
- A **security** researcher
  - Study the security of computer architectures, operating systems, and software applications
    - Find vulnerabilities in CPUs and software
    - Build systems to defend against various attacks
  - **My current research interest**: Build secure systems for confidential computing and AI

Why Are You Here?

- Reason 1: OS is a mandatory course
  - **Yes, you have to pass!**
- Reason 2: Rumor has it: the course is a breeze to pass
  - **No,** it is **notoriously challenging**, which involves complex concepts like process scheduling, memory management, and synchronization. It requires a strong understanding of C/C++ programming and low-level system interactions.
- Reason 3: **Operating system is one of the most important courses for CS majors**
  - **Yes**, it helps you (1) understand how computer systems work, (2) understand abstraction and interfaces, (3) develop system-level thinking, (4) prepare for advanced topics, (5) enhance career opportunities

Content of This Course

- We will study the **theory of operating systems**
  - Virtualization, concurrency, persistence
  - Management of CPU, memory, I/O and storage

- We will learn the **implementation of operating systems**
  - Hands-on experience with an educational operating system (xv6)
  - Run your OS on ~~(emulated)~~ RISC-V CPU ~~(via QEMU)~~

Prerequisites

- **C Programming**
  - All course projects will use C (or assembly)
  - **Self-learning is expected!**
- Data structure and algorithm analysis
  - It helps you understand OS concepts, e.g., LRU
- **Computer Organization**
  - How do computers work?
  - Hardware and software coordination
  - **Sophomore? Intelligence Science and Technology? Self-learning is expected!**

Reference Books

- **Operating Systems: Three Easy Pieces,** Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau
  - Virtualization (Process, scheduling, memory address space, swapping)
  - Concurrency (Threads, locks, semaphores)
  - Persistence (I/O, storage, file systems)
  - **Online available: https://pages.cs.wisc.edu/~remzi/OSTEP/**

- **Slides**
  - Download from BB before each class
  - **Take notes in class**

Course Structure

- **Lectures** (2 hours per week)
  - Get the main ideas and concepts (mostly in English, may repeat in Chinese)
  - lectures will not follow textbooks (but read textbooks will help)
  - Download slides (before class, and check for updates after)
  - Take notes and ask questions
- **Lab** (2 hours per week)
  - Tutorials on kernel code and lab assignments
  - Do the lab exercises / projects
  - All assignments (even written ones) submitted through labs

Gradings

- Lecture participation: 5%
  - Collect 100 points in quizzes to get full points
  - **No need to request leave in the system.**
- Lab participation: 14%
  - In-class assignments
- Assignment: 16%
  - 5 coding/written assignments
  - Pick best 4 out of 5
- Project: 20%
  - Teamwork: 2 - 3 students per group
- Mid-term exam: 20%
  - Apr. 20th, Sunday, 14:00 - 15:50
- Final exam: 25%
  - Closed book (for all exams)
  - No cheating paper (for all exams)

Academic Misconduct

- If an undergraduate assignment is found to be plagiarized, **the first time** the score of **the assignment will be 0**.
- **The second time** the score of **the course will be 0**.
- If a student does not sign the Assignment Declaration Form or cheats in the course, including regular assignments, midterms, final exams, etc., in addition to the grade penalty, the student will not be allowed to enroll in the two CS majors through 1+3, and **cannot receive any recommendation for postgraduate admission exam exemption** and all other academic awards.

Tips: How to Pass

- Come to lectures/lab tutorials
  - Take notes, ask questions, take attendence
  - **Pay attention to quiz and in-class lab assignments**
  - **Exams make a big difference in your final grade**
- Visit Blackboard
  - Announcement
  - Lecture slides (before classes, may update afterwards)
  - Assignments and solutions
- Be ready to code!
  - Finish assignments on time
  - **Spend more time on your project!**

How about ChatGPT/Deepseek/Kimi/Doubao?

OS Concept: Processes

- A process is a program in execution
  - Program is a passive entity and process is an active entity.
- Process needs resources to accomplish its task
  - CPU, memory, I/O, files
  - Process termination requires reclaim of any reusable resources
- Process executes instructions sequentially, one at a time, until completion
  - Single-threaded process has one program counter specifying location of next instruction to execute
  - Multi-threaded process has one program counter per thread
- Typically, system has many processes, some user, some operating system running concurrently on one or more CPUs
  - Concurrency by multiplexing the CPUs among the processes / threads

OS Concept: Process Management

- Creating and deleting both user and system processes
- Suspending and resuming processes
- Providing mechanisms for process synchronization
- Providing mechanisms for process communication
- Providing mechanisms for deadlock handling

OS Concept: Memory

- DRAM (Dynamic Random Access Memory) is the main memory used for all desktop, laptops, servers, and mobile devices
- CPU only directly interacts with the main memory during execution
  - All data in memory before and after processing
  - All instructions in memory in order to execute
- OS manages the main memory for kernel and processes
  - OS dictates which process can access which memory region

OS Concept: Memory Management

- Memory management determines what is in memory when
  - Optimizing CPU utilization and computer response to users
- Memory management activities
  - Keeping track of which parts of memory are currently being used and by whom
  - Deciding which processes (or parts thereof) and data to move into and out of memory
  - Allocating and deallocating memory space as needed

OS Concept: Storage Management

- OS provides uniform, logical view of information storage
  - Abstracts physical properties to logical storage unit - file
  - Each medium is controlled by device (i.e., disk drive, tape drive)
    - Varying properties include access speed, capacity, data-transfer rate, access method (sequential or random)
- File-System management
  - Files usually organized into directories
  - Access control on most systems to determine who can access what
  - OS activities include
    - Creating and deleting files and directories
    - Primitives to manipulate files and dirs
    - Mapping files onto secondary storage
    - Backup files onto stable (non-volatile) storage media

OS Concept: I/O Subsystem

- One purpose of OS is to hide peculiarities of hardware devices from the user
- I/O subsystem responsible for
  - Memory management of I/O including
    - buffering (storing data temporarily while it is being transferred)
    - caching (storing parts of data in faster storage for performance)
  - General device-driver interface
  - Drivers for specific hardware devices

OS Concept: Protection and Security

- **Protection** – any mechanism for controlling access of processes or users to resources defined by the OS
- **Security** – defense of the system against internal and external attacks
  - Huge range, including denial-of-service, worms, viruses, identity theft, theft of service
- OS determines which users can do what
  - User identities (user IDs, security IDs) include name and associated number, one per user
  - User ID then associated with all files, processes of that user to determine access control
  - Group identifier (group ID) allows set of users to be defined and controls managed, then also associated with each process, file
  - Privilege escalation allows user to change to effective ID with more rights

Announcement

- No lecture on Feb 25th (next week)
  - Conference trip
- Labs are not affected
  - Ms. Yun Shen will take my place on Feb 25th.

Thank you!

